

Lijie Ding

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For more: [Github](#), [LinkedIn](#), [Google Scholar](#), [CV](#)

Skills

AI/ML: Agentic AI, Large Language Models (LLMs), RAG, Physics-informed Machine Learning, Deep Learning, Bayesian inference, CNNs, VAEs, Gaussian Processes

AI Frameworks: PyTorch, CrewAI, Agent SDK, MCP, scikit-learn, Claude Code, OpenCode

Scientific Computing: Monte Carlo (MC), Molecular Dynamics (MD), LAMMPS, HOOMD-blue, HPC, Slurm

Programming: Python, C/C++, NumPy, Pandas, SQL, Git

Experience

Oak Ridge National Laboratory (ORNL)

Postdoctoral Research Associate

Oak Ridge, TN

May 2024 – Present

Agentic AI for Science: Built multi-agent systems (ColPackAgent, ToPolyAgent, SasAgent) to orchestrate complex scientific workflows such as small-angle scattering data analysis, MD, and MC simulations end-to-end from natural-language prompts, with no coding or simulation expertise required.

LLM Application: Architected an LLM-driven proposal review pipeline using pairwise preference and AI-as-a-judge, achieving 800× cost reduction while maintaining review performance for ORNL SNS, validated on 2,618 historical proposals over 15 years.

AI for Characterization: Developed and trained physics-informed ML/DL models in PyTorch (e.g., Gaussian process regression, Bayesian inference, VAEs, CNNs) to extract hidden information from SAS, applied to colloids, polymers, lamellae, and polydisperse systems, yielding 12 journal publications (7 first-author).

Scientific Computing: Developed MD/MC simulations on HPC clusters (Slurm) using C++, Python, LAMMPS, and HOOMD-blue for colloids, polymers, active fluid and rheology systems.

Goldman Sachs

Quantitative Strategist, Vice President (Jan24–May24), Associate (Jun22–Dec23)

New York, NY

Jun 2022 – May 2024

Model Development: Owned the full lifecycle development of production valuation models, ensuring high reliability and performance for interest rate (IR) derivative pricing. Executed the SOFR transition for IR products ahead of the Jun 2023 cessation deadline, and conducted statistical validation for pricing.

Large-Scale Attribution Analysis: Built analysis pipelines to explain price variance vs. market consensus and decompose moves into key risk factors for multi-billion-dollar inventory of IR instruments, used by desk strategists and trader to investigate pricing moves.

Brown University

Ph.D. Research Assistant

Providence, RI

Sep 2017 – Jun 2022

Scientific Computing: Developed high-performance MC simulations for colloidal membranes in C++. Performed mathematical modeling and numerical studies in Python.

Scientific Data Analysis: Developed OpenCV-based image analysis tools for DNA Brownian motion and performed time series analysis.

Publications (27 Journal Articles, 1 Conference Proceeding, 4 Preprints)

Lijie Ding et al., “ToPolyAgent: AI Agents for Coarse-Grained Bead-Spring Topological Polymer Simulations,” *Digital Discovery* **5**, 901–909 (2026). DOI: [10.1039/D5DD00471C](https://doi.org/10.1039/D5DD00471C)

Lijie Ding et al., “LLMs Can Assist with Proposal Selection at Large User Facilities,” (2025). arXiv: [2512.10895](https://arxiv.org/abs/2512.10895)

Lijie Ding et al., “Machine Learning-Assisted Profiling of a Kinked Ladder Polymer Structure using Scattering,” *Digital Discovery* **4**, 1570–1577 (2025). DOI: [10.1039/D5DD00051C](https://doi.org/10.1039/D5DD00051C)

H. Du, J. Su, J. Li, **Lijie Ding**, et al., “Which LLM Multi-Agent Protocol to Choose?,” *Proceedings of the International Conference on Machine Learning (ICML)* (2026). OpenReview: [GZRiXE2zBe](https://openreview.net/forum?id=GZRiXE2zBe)

More: full list can be found at [Google Scholar](#), or [CV](#)

Education

Brown University

Ph.D. in Physics (Dissertation: Chiral Liquid Crystals on Deformable Surfaces)

Providence, RI

2017 – 2022

University of Science and Technology of China (USTC)

Hefei, China

B.Sc. in Applied Physics

2013 – 2017